

Time Value of Money Problems

Problem 1. Consider the following amount function, which is a function of the number of years elapsed, t :

$$A(t) = Xt^2 + Yt + Z.$$

You are given that:

- $A(0) = 500$
- $A(1) = 521$
- $A(4) = 596$

Calculate the future value at time 5 of \$600 that is deposited at time 0.

Problem 2. Fund A accumulates at a force of interest,

$$\delta_t^A = \frac{0.05}{1 + 0.05t}, \quad t \geq 0.$$

Fund B accumulates at an annual force of interest of 0.05.

You are given that:

- The amount in Fund A at time 0 is 1,000.
- The amount in Fund B at time 0 is 500.
- The amount in Fund C at any time t is equal to the sum of the amount in Fund A and the amount in Fund B.
- Fund C accumulates at a force of interest, δ_t^C , $t \geq 0$.

Calculate δ_2^C .

Problem 3. Clayton has two savings accounts. Savings account #1 earns interest at an annual effective discount rate of 4.7619%, and savings account #2 earns interest at a nominal annual interest rate compounded monthly of 11.3861%. Clayton has not made any deposits or withdrawals since March 1, 2014, when the amount in savings account #1 was five times the amount in savings account #2.

The sum of the two accounts on March 1, 2025 was \$130,000.

Calculate the sum of the two accounts on March 1, 2014.

Problem 4. Three deposits are made into a fund as follows:

- A deposit of \$100 now
- A deposit of \$300 in one year
- A deposit of \$500 in two years

An annual simple interest of 6% is credited in the first and second year. For the third year, interest is credited at

$$\delta_t = 0.06t,$$

where t is the number of years from now.

Find the accumulated value of the fund at the end of three years.

Problem 5. The present value of \$200 paid at the end of n years, plus the present value of \$100 paid at the end of $2n$ years, is \$200.

Determine the annual effective rate of interest.

A. $\left(\frac{\sqrt{3}+1}{2}\right)^{1/n} - 1$

B. $1 - \left(\frac{\sqrt{3}-1}{2}\right)^{1/n}$

C. $\left(\frac{\sqrt{3}-1}{2}\right)^{1/n} - 1$

D. $\left(\frac{\sqrt{3}+1}{2}\right)^{1/(2n)} - 1$

E. $\left(\frac{\sqrt{3}-1}{2}\right)^{1/n} + 12$

Problem A. \$1,000 is deposited into a savings account at time $t = 0$. No other amounts are deposited. The accumulated amount in the account at any time is given by

$$A(t) = 1000 \left(1 + \frac{2t}{35} \right)^2.$$

At what time t is the force of interest equal to 0.04?

Problem B. Joe deposits \$10 today and another \$30 in five years into a fund paying simple interest of 11% per year.

Tina will make the same two deposits, but the \$10 will be deposited n years from today and the \$30 will be deposited $2n$ years from today. Tina's deposits earn an annual effective interest rate of 9.15%.

At the end of 10 years, the accumulated amount of Tina's deposits equals the accumulated amount of Joe's deposits.

Calculate n .

Problem C. Jim deposits \$100 into an account that credits:

- An effective rate of discount of 3% for the first year;
- An effective rate of discount of 6% for the second year;
- An effective rate of discount of 9% for the third year; and
- $\delta_t = 0.02t$ for the fourth and fifth years.

How much is in the account at the end of the fifth year?
